

Detecting Resistance, Demonstrating Utility

*A critical analysis of nanopore sequencing, minority resistance signals, and clinical inference in a *Klebsiella pneumoniae* case study*

A critical analysis of "Detection of hidden antibiotic resistance through real-time genomics"

Sauerborn, Corredor, Reska, Perlas, Atum, Goldman, Wantia, Prazeres da Costa, Foster-Nyarko & Urban (2024), Nature Communications, 15, 5494

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1. Central Claim and Evidentiary Overview

Sauerborn et al. use a single multidrug-resistant *Klebsiella pneumoniae* infection to argue that real-time nanopore genomics can detect low-abundance, plasmid-borne antibiotic resistance that routine clinical diagnostics miss, and that this capability could plausibly improve treatment decisions. The strongest, best-supported part of that claim is narrow: sequencing detected a low-abundance blaKPC-14-labelled signal in the pre-treatment isolate despite its reported phenotypic CAZ-AVI susceptibility, and in the post-treatment isolate, routine testing detected CAZ-AVI resistance while the carbapenemase assay returned a negative result. Everything built on top of that observation — that this approach would have changed treatment, that it generalises beyond one patient, or that it is ready for bedside use — is inference rather than demonstrated result.

This analysis is organised around one question: what does this retrospective case actually demonstrate about detecting low-abundance resistance signals, and what would still need to be shown before that detection translates into reliable, timely clinical decision-making? The article is a proof-of-concept case study, not a validation study, and the two kinds of evidence should not be read as interchangeable.

Evidentiary overview. One patient; two cultured *K. pneumoniae* isolates (endotracheal aspirate, then blood culture); routine diagnostics performed in real time and used to guide treatment; nanopore sequencing performed retrospectively, after the clinical course had concluded, on stored isolates that were thawed and regrown; the article does not report the sequencing results as having influenced any treatment decision.

2. Theoretical Foundation

Antibiotic resistance is one of the ten global health threats named by the World Health Organization, and delayed or inappropriate antibiotic therapy is associated with higher mortality in resistant infections. The standard diagnostic path — culture, subculture, phenotypic susceptibility testing — can take one to several days before a clinician learns which drugs will work, and phenotypic testing measures whether bacteria grow in the presence of a drug rather than which gene is responsible. Genotypic resistance prediction instead sequences bacterial DNA and looks for known resistance genes, such as the blaKPC family of genes that encode *K. pneumoniae* carbapenemase (KPC) enzymes. KPC enzymes break down carbapenem antibiotics, a drug class often reserved for serious infections, so their presence or absence has direct treatment implications.

Minimum inhibitory concentration (MIC) is the lowest drug concentration that visibly stops bacterial growth in a lab test; it is the standard measure behind a phenotypic susceptibility result. A gene and its protein product are not the same thing — blaKPC is the gene; KPC is the enzyme it encodes — and the article's sequencing workflow detects gene sequences, not enzyme activity directly.

Nanopore sequencing (Oxford Nanopore Technologies) reads long stretches of DNA in real time as single molecules pass through a protein pore, producing a current signal that is translated (“basecalled”) into nucleotide sequence. Reads can be analysed as they arrive rather than only after a full run completes, and because sequencing effort can be extended for as long as a sample keeps producing usable DNA, the technology lends itself to what this article calls an “adaptive” approach: sequence until a given confidence threshold is reached. This is a different sense of “adaptive” from nanopore’s selective read-rejection feature (sometimes also called adaptive sampling), which this study did not use.

A resistance gene present in only a small fraction of a bacterial population, or at low copy number relative to the chromosome, is a low-abundance signal. Detecting it depends on sampling enough DNA molecules, and by chance a small sample can miss a rare variant even if it is genuinely present — a form of sampling uncertainty distinct from any error in the sequencing chemistry itself. A technical replicate repeats a measurement on the same biological source (here, the same stored isolate, regrown and resampled) and is informative about measurement

consistency; it is not a biological replicate, which would come from an independent patient or independent infection and would speak to generalisability.

3. Data and Study Design

This is a retrospective case study of one immunocompromised patient treated at the University Hospital rechts der Isar in Germany. Two *K. pneumoniae* isolates from that patient were cultured under routine clinical care:

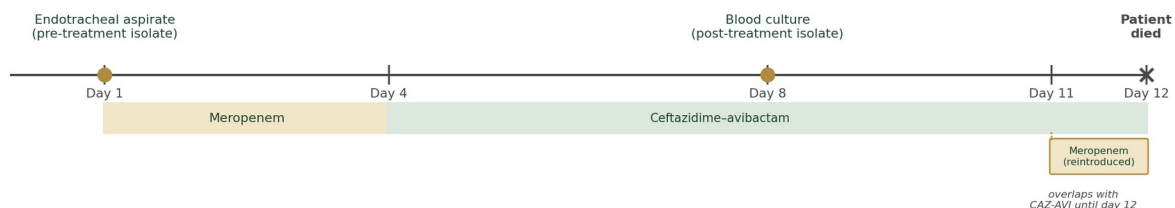
- The pre-treatment isolate, cultured from an endotracheal aspirate on day 1, when the patient presented with fever.
- The post-treatment isolate, cultured from a blood culture on day 8, after clinical deterioration under ceftazidime–avibactam (CAZ-AVI) therapy.

Both isolates went through routine phenotypic and immunochromatographic diagnostics in real time, and those results directly informed the patient's treatment. After routine diagnostics were complete, both isolates were stored at -80°C , later thawed, regrown on agar, and sequenced. The sequencing described throughout the article, including its framing as “real-time genomics,” was therefore a retrospective simulation of how such a workflow could have been used, not a prospective diagnostic step in this patient's actual care. The retrospective sequencing was not reported as guiding treatment.

A second, separate sequencing experiment extended this picture: the pre-treatment isolate was resubcultured a second time to generate a technical replicate, which was then sequenced for a further 8 hours to illustrate how extending a sequencing run might accumulate more evidence. This technical replicate is still the same patient and the same original isolate — resampled and resequenced — not a second clinical case or an independent biological sample.

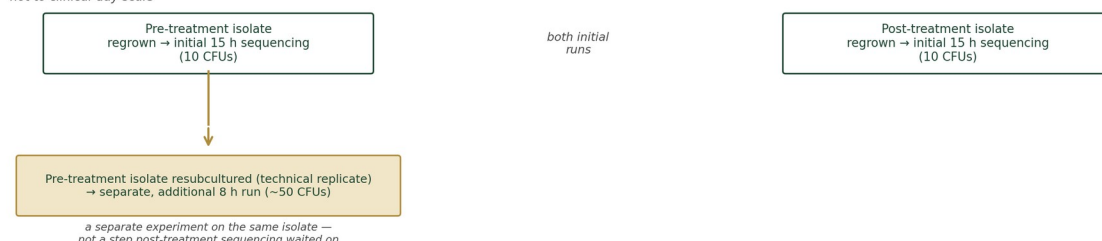
Constructed for this analysis — not reproduced from the article

Clinical course (days, actual patient timeline)



Retrospective sequencing work (performed after routine diagnostics; not reported as guiding treatment)

not to clinical-day scale



Both isolates were first sequenced for 15 h. The pre-treatment isolate's technical replicate was separately resequenced for a further 8 h; the post-treatment isolate's 15 h run did not depend on it.

Figure 1. Two timelines: the actual clinical course and the retrospective sequencing work. Conceptual diagram constructed for this analysis. Clinical timeline follows the day markers in the article's Figure 2: meropenem was reintroduced at approximately day 11, continuing alongside CAZ-AVI until day 12. Both initial 15 h sequencing runs were performed independently; the additional 8 h run was a separate experiment on a resubcultured pre-treatment technical replicate and did not precede post-treatment sequencing.

A further design point matters for interpreting change over time: the two isolates differ in both collection day and anatomical source (respiratory tract aspirate vs. blood). Any difference observed between them could reflect a true change in the infecting population, a difference between anatomical niches, or an artifact of separate culturing and sampling — the study design cannot fully separate these explanations.

The clinical sequence, stated without embedding treatment recommendations: empirical meropenem was started; routine diagnostics reported carbapenem resistance with KPC suspected and CAZ-AVI susceptibility, prompting a switch to CAZ-AVI; the patient improved and then deteriorated; the post-treatment isolate showed CAZ-AVI resistance together with restored in-vitro carbapenem susceptibility, and the routine carbapenemase assay did not detect a carbapenemase in this isolate; meropenem was reintroduced; the patient died. The article does not claim, and this analysis does not claim, that undetected resistance caused the death or that sequencing would have prevented it.

4. From Retrospective Detection to Clinical Utility — The Central Critique

The article's central scientific contribution combines two distinct findings, and separating them clarifies what each demonstrates. Sequencing detected a low-abundance blaKPC-14-labelled signal in the pre-treatment isolate despite its reported phenotypic CAZ-AVI susceptibility. In the post-treatment isolate, routine testing detected CAZ-AVI resistance, but the carbapenemase assay returned a negative result, while sequencing of that isolate identified numerous copies of the same blaKPC-14-labelled gene. In the post-treatment isolate, sequencing supplied mechanism-level information that the carbapenemase assay did not provide, although phenotypic testing detected CAZ-AVI resistance. The low-abundance pre-treatment signal is consistent with, but does not prove, the authors' selection hypothesis; the carbapenemase assay was not shown to have failed at that earlier time point. The question this section takes up is how far these findings travel toward the discussion's broader claims about clinical decision-making and patient outcomes.

Five distinct claims get conflated in the article's framing, and each requires separate evidence:

- A validated diagnostic threshold — a pre-specified copy-number or confidence level, tested against known-resistant and known-susceptible samples, above which a clinician should act. The article reports what was observed in one case, not a threshold derived from a validation set.
- Superior sensitivity and specificity across patients — requires many patients, ideally with a reference standard, to estimate error rates. One case cannot estimate either quantity.
- Reliable detection directly from clinical specimens — the study sequenced cultured, regrown isolates, not DNA extracted directly from aspirate or blood. The authors themselves flag this as a limitation (Section 9).
- A prospectively delivered clinical result — sequencing here was retrospective and simulated; no result was delivered to a clinician during the patient's care.
- Improved treatment or survival — the patient in this case study died. The article's claim that earlier detection “would have” improved management is a counterfactual, not an observed outcome.

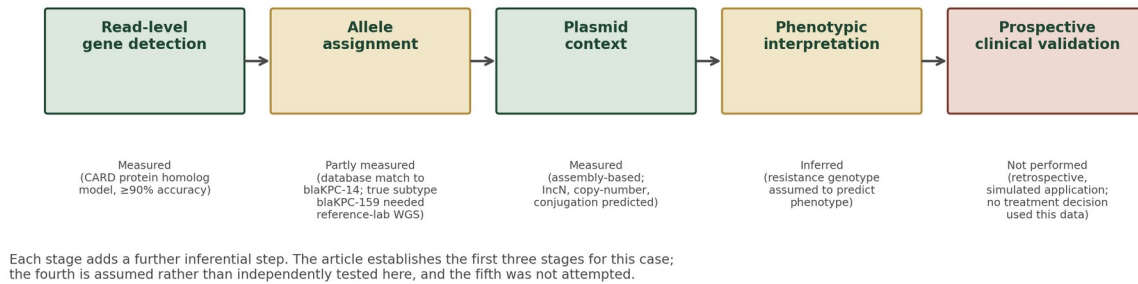


Figure 2. *From sequence signal to clinical evidence.* Conceptual diagram constructed for this analysis, separating what the study measured from what it assumed or did not attempt.

None of this makes the finding unimportant. A proof-of-concept study is allowed to stop short of clinical validation, and the authors are reasonably explicit in their Discussion that “further research is needed.” The critique here is about the gap between that qualified language and the more sweeping claims elsewhere in the same Discussion — for instance, that real-time genomics “would have surpassed current clinical practice” — which reads as a stronger claim than a single retrospective case can support.

5. Sequencing and Bioinformatics Workflow

DNA was extracted from regrown isolates (10 colony-forming units, or CFUs, per isolate in the first round; roughly 50 CFUs from a second subculture of the pre-treatment isolate for the additional 8-hour run) using an automated magnetic-bead protocol. Libraries were prepared with the SQK-RBK004 rapid barcoding kit and sequenced on a portable MinION Mk1b device with R9.4.1 flow cells.

Two analytical tracks ran on the resulting reads, and the article’s own framing distinguishes them; conflating them would overstate how much assembly the rapid detections actually required.

Rapid, read-level track (no finished assembly required). Guppy v6.3.2 high-accuracy basecalling converted raw signal to sequence; Porechop and NanoFilt trimmed adapters and removed low-quality ($Q < 9$) or short (< 200 bp) reads; the EPI2ME Fastq Antimicrobial Resistance workflow then ran taxonomic classification (WIMP, using NCBI RefSeq via Centrifuge) and resistance-gene detection directly against reads using the Comprehensive Antibiotic Resistance Database (CARD) protein homolog model, the most conservative of CARD’s detection models. Detections below 90% accuracy under this model were discarded. This is the track that produced the early, cumulative gene-copy counts referenced throughout the article, including in Table 1 and Figure 3.

Later, assembly-based track. Flye assembled the filtered reads de novo into a chromosome and plasmid contigs, polished with Minimap2 and Racon; SAMtools depth measured coverage; Pathogenwatch (integrating Kleborate) performed species-complex assignment, virulence and resistance marker annotation, and core-genome multilocus sequence typing (cgMLST); Bandage visualised assembly graphs; MOB-suite annotated plasmid features (relaxase type, origin of transfer, mate-pair formation type) and estimated conjugative potential, visualised through mobileOG-db in ProkSee. This track produced the plasmid-context findings discussed in Sections 7 and 8 below — plasmid copy number, IncN plasmid identity between isolates, and predicted conjugative machinery.

Software versions matter here mainly for reproducibility rather than for evaluating the finding itself: this analysis assesses the CARD protein homolog model and Flye/MOB-suite pipeline actually used, rather than substituting newer basecalling models or assembly tools that might perform differently.

6. What Was Actually Compared with Routine Diagnostics?

A common simplification is to describe the routine workflow as “VITEK2 testing,” but the Methods describe four distinct routine tests, each with a different job, and the article's central claim rests on which specific test failed to detect which specific thing.

Test	Purpose	Sample input	Relevant limitation
MALDI-TOF MS	Species identification	Single CFU	Not a resistance test; not implicated in the missed detection.
VITEK2 AST-GN69 card	General MIC panel, gram-negative susceptibility profile	Up to 3 CFUs, 0.5 McFarland suspension	Does not itself detect carbapenemases or measure CAZ-AVI MIC — those require the two tests below.
O.K.N.V.I. Resist-5 lateral flow assay	Carbapenemase detection (OXA-48, NDM, VIM, IMP, and the most prevalent KPC subtypes, e.g. KPC-2/3)	Colony-derived	Targets known, prevalent subtypes; a novel or atypical KPC variant may not be recognised. This is the test that returned a negative carbapenemase result for the post-treatment isolate.
Liofilchem CAZ-AVI MIC strip	CAZ-AVI-specific MIC measurement	Up to 3 CFUs, 0.5 McFarland suspension	Phenotypic only; reports susceptibility category, not mechanism.
Nanopore sequencing (this study)	Genotypic resistance-gene detection and plasmid context	10 CFUs (initial); ~50 CFUs (additional pre-treatment run)	Retrospective, on cultured isolates; not delivered as a clinical result in this case.

Table 1. What the two workflows measured — purpose, sample input, and limitations of each diagnostic component.

Two points follow directly from this breakdown. First, the routine workflow did successfully detect the post-treatment isolate's CAZ-AVI resistance phenotypically (via the Liofilchem strip); what it failed to detect was the carbapenemase mechanism behind that resistance, via the lateral flow assay. It is inaccurate to describe routine diagnostics as having “missed resistance” at both time points — the miss was specifically a mechanism-level, not phenotype-level, gap in the second isolate.

Second, the comparison is not sampling-matched, though only partly in ways the article's Methods pin down precisely. The VITEK2 panel and the Liofilchem CAZ-AVI MIC strip each used up to three CFUs; the initial sequencing used DNA from ten CFUs; the additional pre-treatment sequencing experiment used roughly fifty CFUs after a further subculture. The Methods do not state a specific colony input for the O.K.N.V.I. lateral-flow carbapenemase assay, so its sampling cannot be compared to sequencing with the same precision. A low-abundance variant is, in principle, easier to catch with a larger effective sample of bacterial DNA. This does not mean the sequencing finding is an artifact of sample size — the article does not test that possibility either way — but the difference in colony input across the tests that do report one is a genuine possible influence sitting alongside the difference in measurement technology, and a fair reading should hold both explanations open rather than attributing the entire gap to nanopore sequencing's superior sensitivity.

7. Low-Abundance Detection, Variant Identity, and Quantification

KPC-14 detection versus KPC-159 identification

This distinction is the single most important technical nuance in the article, and it is easy to flatten in a summary. The nanopore workflow's CARD-based classifier labelled the resistance signal in both isolates as blaKPC-14, an existing, previously characterised database entry. It was only when the post-treatment isolate was independently sent to the German National Reference Center for Gram-negative bacteria, and sequenced there by short-read whole-genome sequencing (Illumina MiSeq), that the variant was identified as a previously undocumented KPC subtype, subsequently registered as blaKPC-159 (NCBI accession OQ450354.1). A BLASTn comparison then showed 875 of 876 nucleotides identical (about 99.9%) between blaKPC-159 and the nanopore workflow's blaKPC-14 query sequence.

Both readings of this result are defensible and worth holding together. In the workflow's favour, a near-identical match to a known resistance-associated allele was enough to flag a clinically relevant signal even though the pipeline could not, on its own, name the exact novel subtype — flagging “resistance-associated gene present” does not require allele-perfect identification to be useful. Against that, the original real-time pipeline did not, and could not, report “KPC-159” at the time; that identification depended on a separate short-read reference-laboratory step applied only to the post-treatment isolate. Nothing in the article indicates that the pre-treatment isolate's single low-abundance blaKPC-14-labelled read was ever independently confirmed by short-read sequencing as KPC-159 specifically — only the post-treatment isolate went to the reference centre. The pre-treatment signal's precise allele identity is therefore inferred by similarity, not independently established. The article also reports no cloning, enzyme kinetics, or isogenic strain testing to functionally confirm what KPC-159 does; its behaviour is inferred from sequence similarity to characterised KPC variants, not measured directly here.

Three meanings of “copy number”

The article uses related but distinct quantities, and treating them as interchangeable would misrepresent what was measured:

- Resistance-gene detections reported by the sequencing workflow — a count of read-level resistance-gene detections that the CARD protein homolog model classified as matching a resistance gene above the 90% accuracy threshold (Table 1's “CN” column).
- Plasmid copy-number estimates — the ratio of sequencing depth on a plasmid contig to sequencing depth on the chromosomal contig, giving an estimated 3 (pre-treatment) and 4 (post-treatment) IncN plasmid copies per chromosome.
- Normalised resistance-gene abundance — the blaKPC-14 copy-number estimate divided by the copy-number estimate of the most abundant resistance gene on the same plasmid (blaTEM-4), reported as rising from 0.56% to 26.6% between isolates.

None of these numbers is a direct count of plasmids per bacterium, nor a fraction of infected cells, nor a directly measured within-patient resistant-cell frequency. The read-level detection counts reflect how many reads passed the CARD workflow's matching and accuracy criteria; the plasmid copy-number and normalised abundance figures are sequencing-depth-derived estimates, sensitive to culture conditions, DNA extraction efficiency, library preparation, sequencing coverage, and the normalisation choice itself. The 0.56%-to-26.6% figure is a useful relative signal of enrichment, but the article does not test how much of that change reflects true biological enrichment during treatment versus culture-step or extraction variability between the two separately processed isolates.

Evidence	What was measured	Interpretive caveat
Pre-treatment single blaKPC-14 detection	1 read-level detection, 92.6% accuracy, 15 h run	Below the study's own working threshold for a confident call; not independently allele-confirmed.
Additional 8 h technical-replicate run	4 further high-accuracy detections; first additional copy detected ~2 h in	Same isolate, resubcultured with ~50 CFUs — not a second patient or an uninterrupted extension of the same library.
KPC-159 identification	875/876 nucleotide match (99.9%) to blaKPC-14 query, via reference-centre short-read WGS	Performed only on the post-treatment isolate; not a real-time nanopore call.
Normalised blaKPC-14 abundance change	0.56% → 26.6% (relative to blaTEM-4 on the IncN plasmid)	A depth-derived relative estimate, not a measured fraction of resistant cells.
IncN plasmid copy number	3 (pre-) → 4 (post-treatment), relative to chromosome coverage	Depth-ratio estimate; culture and extraction effects not separately quantified.

Table 2. Evidence and interpretation — what each reported quantity does and does not establish.

Detection counts across the two runs

The authors report one blaKPC-14-labelled detection in the initial 15-hour run and four additional detections in the separate 8-hour technical-replicate run. Figure 3 distinguishes the initial detection from the four additional detections. These are sequencing detections across two experiments, not five independently sampled resistant cells.

8. Timing, Sampling, and the Selection Hypothesis

Comparing timing on equal terms

The article's workflow schematic reports roughly 1.5 hours to first sequencing data (DNA extraction, library preparation, and the start of sequencing and automated analysis), against up to 52 hours for the routine workflow after subculturing. That comparison describes time to first data, not time to a validated resistance result.

Distinguishing the steps matters:

- Specimen collection, followed by initial culture and subculture (both workflows require this before any isolate-based analysis begins).
- DNA extraction and library preparation (~1.5 h in the nanopore workflow).
- First sequencing data and first resistance-associated hit — available early, but based on a single or few reads.
- Additional evidence supporting the authors' resistance prediction — obtained during the separate 8-hour technical-replicate run; this did not establish a validated clinical reporting threshold.
- Reference-laboratory confirmation was performed on the post-treatment isolate; prospective reporting of nanopore findings to guide treatment was not demonstrated.

The two-hour figure that recurs in the article's framing — a second blaKPC-14 copy detected within two hours of the additional run — describes two hours into a second sequencing experiment on a resubcultured technical replicate that itself followed an earlier 15-hour run and a fresh round of culturing, DNA extraction, and library preparation. It is not two hours from the original specimen collection, and the second run also used a larger colony

input (~50 vs. 10 CFUs), so it is not simply an uninterrupted continuation of the same sequencing library. Figure 3 in the original article, which plots cumulative detections against time for this additional run, is informative about within-run accumulation of evidence but should not be read as a specimen-to-result turnaround time.

The selection hypothesis

The Discussion proposes that a low-abundance blaKPC-14-carrying IncN plasmid was already present at low levels in the pre-treatment infection and became dominant under CAZ-AVI selective pressure, explaining the switch from carbapenem resistance/CAZ-AVI susceptibility to carbapenem susceptibility/CAZ-AVI resistance between isolates. This is a reasonable and testable hypothesis, and it is consistent with what was observed: both isolates were assigned to the same high-risk sequence type (ST147); the IncN plasmids carrying blaKPC-2 (pre-treatment) and blaKPC-14 (post-treatment) were about 99.7% identical by sequence alignment and shared key mobilisation features (relaxase type MOB_F, mpf type MPF_T, oriT type MOB_F), and were predicted, based on those features, to be conjugative; and no close relatives were found within the standard 50-allele cgMLST clustering threshold in the comparison collection queried.

Consistency with a hypothesis is not proof of it, and several specific claims go beyond what one patient, two time points, two different specimen sources, and culture-based sampling can establish:

- The exact resistant fraction present in the original infection — not measured directly; only inferred from post-culture sequencing depth.
- Plasmid carriage in specific individual bacterial cells — sequencing here reports population-level averages, not single-cell data.
- Direct horizontal gene transfer occurring during treatment — the IncN plasmids were predicted to be conjugative based on their genetic machinery; conjugation itself was not observed or experimentally demonstrated.
- A specific mutation or selection event at a known time — the data show two snapshots, not a time-resolved evolutionary trajectory.
- That all observed differences arose inside the patient, rather than being partly shaped by which colonies were picked, cultured, and sequenced at each time point.
- Global uniqueness of this strain — the absence of close relatives in the specific comparison collection queried does not establish that no close relatives exist worldwide, only that none were found in that dataset.

9. Strengths and Limitations

Read fairly, the study earns credit on several fronts. It detected a resistance-associated signal that a purpose-built carbapenemase assay did not flag, using a portable, comparatively low-cost sequencing platform. It connected that gene-level finding to plasmid-level context (identity between isolates, copy-number estimates, predicted mobility) rather than stopping at gene presence or absence. It demonstrated, through the additional 8-hour run, one concrete way of accumulating more evidence when an initial signal is uncertain. It compared molecular findings against a real, documented clinical course rather than a purely computational benchmark. It obtained independent reference-laboratory characterisation of a genuinely novel KPC subtype. And it released both sequencing data (NCBI BioProject/SRA PRJNA1041345) and analysis code, supporting attempts at reproduction.

The authors are also reasonably candid about two limitations of their own work: that focusing on cultured isolates rather than direct metagenomic sequencing of patient specimens may have missed non-culturable organisms contributing to the infection, and that two isolates from one patient cannot capture how plasmid selection evolved

with any temporal resolution, a constraint the retrospective design made unavoidable, since sample size and direct specimen access could not be adjusted after the fact.

Strength	Corresponding limitation	What a next study would need to test
Detected a signal routine diagnostics missed	One case; cannot estimate sensitivity or specificity	A multi-patient cohort with a reference standard
Linked gene findings to plasmid context	Copy-number estimates are depth-derived, not direct counts	Orthogonal quantification (e.g. digital PCR) of plasmid abundance
Extended sequencing accumulated more evidence	No pre-specified, validated confidence threshold	A prospectively defined decision rule, tested against known outcomes
Reference-lab confirmation of a novel subtype	Confirmation applied to one isolate, not the low-abundance pre-treatment read	Short-read confirmation of low-abundance calls specifically
Data and code publicly released	Retrospective on cultured isolates, not direct specimens	A metagenomic, direct-from-specimen comparison arm

Table 3. Strengths, limitations, and the tests that would address each limitation.

A fair overall read is that the study demonstrates a real, useful instance of retrospective detection with credible mechanistic context, while prospective clinical utility and general diagnostic superiority over established methods remain open questions rather than settled ones.

10. Reproducibility and Data Availability

The article reports its sequencing data as deposited at NCBI under BioProject/SRA submission PRJNA1041345, the blaKPC-159 sequence under NCBI accession OQ450354.1, and its analysis scripts at a versioned GitHub release (Genomics4OneHealth/AMR_nanopore, tag v0). This analysis reports what the article states about these resources; it has not independently retrieved the deposited sequencing reads, executed the released code, or re-run the CARD, Flye, or MOB-suite pipeline steps described in the Methods. That distinction — between an article reporting that data and code are available and independent verification that they reproduce the stated results — matters, since open availability supports reproducibility attempts without guaranteeing that an independent run would return identical gene-detection counts, assembly structures, or copy-number estimates, particularly given how sensitive the reported quantities are to software versions and default parameters.

The Methods report specific versions throughout (Guppy v6.3.2, Porechop v0.2.3, NanoFilt v2.8.0, the EPI2ME Fastq Antimicrobial Resistance workflow v2023.04.26-1808834, Flye v2.9.1, Minimap2 v2.18, Racon v1.5, SAMtools v1.19.2, Pathogenwatch v2.3.1, Bandage v0.90, MOB-suite v3.1.8, SeqKit v2.8.0), which is good practice for reproducibility; a later reader attempting to reproduce this analysis should expect that newer versions of the same tools, particularly the CARD database or basecalling models, could shift the exact accuracy percentages and copy-number estimates reported in Table 1.

11. My Critical Perspective

Reading this piece, the part I find most persuasive is also the most modest: a single low-abundance gene copy, detected on a first 15-hour run, that would not on its own have supported a confident CAZ-AVI resistance call, but that pointed toward exactly what a further 8 hours of sequencing on a resubcultured replicate then confirmed. That is a clean, legible demonstration of how extended, adaptive sequencing could work in principle. It does not need the surrounding rhetoric about outperforming clinical practice to be worth reporting.

What I find less persuasive is how quickly the Discussion moves from that single case to language about clinical decision-making and patient outcomes. The sentence describing how real-time genomics “would have surpassed current clinical practice in accurately informing clinical management” is a counterfactual built on a study that was not reported as delivering a result to a clinician. I do not think that sentence is dishonest — the qualifying “would have” is doing real work — but it sits close enough to a factual claim that a reader skimming the Discussion could walk away thinking clinical superiority had been shown rather than proposed.

The sampling asymmetry in Section 6 also nags at me more than the article addresses. Comparing a method with an established three-colony input — VITEK2 and the CAZ-AVI MIC strip — against one that used ten, and then again against one that used fifty, is comparing methods and comparing inputs at the same time. I would not go as far as claiming the whole finding is a sampling artifact — there is no basis in the article for that claim either — but I would have liked to see the authors at least gesture at what happens if the CAZ-AVI MIC strip were run on a larger colony pool, since that would help separate the technology difference from the input-size difference, even if it could not fully isolate the two.

Finally, I keep returning to the KPC-14/KPC-159 naming issue as the part of this paper that rewards a close, second read. It would be easy to write a summary of this article that says “nanopore sequencing detected KPC-159,” and that sentence is not quite true of the real-time workflow as it actually ran. The workflow detected something that matched an existing database entry closely enough to be clinically informative, and a separate process later gave that something its correct name. Both halves of that story are worth keeping, and I think the article's own framing sometimes lets the two blur together more than the Methods section actually supports.

12. What Would Change My Mind

None of the critique above is a claim that the underlying idea is wrong — only that this particular case study cannot yet carry the weight the Discussion places on it. Specific evidence that would shift my assessment toward the stronger clinical claims:

- A pre-specified, validated detection threshold (e.g. a minimum copy-number or confidence score) tested prospectively against a panel of known-resistant and known-susceptible isolates, with reported sensitivity and specificity.
- A multi-patient cohort, ideally with standardised colony input across sequencing and phenotypic methods, so that technology and sampling differences are not confounded.
- At least one case in which a genomically detected low-abundance signal was reported to clinicians in real time and is shown to have influenced an actual prescribing decision, with outcomes tracked.
- Direct-from-specimen sequencing (metagenomic, not culture-based) achieving comparable detection of low-abundance resistance genes, addressing the non-culturable-organism limitation the authors themselves raise.
- Independent short-read confirmation of low-abundance nanopore calls specifically — not only of high-abundance post-treatment findings — to establish that single-copy detections are reliably allele-accurate rather than only reliably present.

Short of that evidence, I read the article as a well-executed demonstration that the underlying detection capability exists, sitting a meaningful distance from a demonstration that it currently changes outcomes.

13. Potential Research Extensions

Extensions within reach of computational reanalysis

1. Time-binned reanalysis of accumulating evidence. Question: how quickly, and how consistently, does detection support for the blaKPC-14-labelled signal accumulate as sequencing time increases? Data: the publicly deposited nanopore reads (PRJNA1041345) from the additional 8-hour run. Method: bin reads by acquisition time and recompute CARD protein homolog model detections and confidence scores at each bin, extending the single cumulative-count curve in the article's Figure 3 into a fuller evidence-accumulation profile. This would describe how quickly supporting detections build up, not establish actual detection accuracy, which would require comparison against an independently validated reference. Limitation: this would still be one isolate's replicate run, so any threshold derived from it would need external validation before generalising.

2. Discriminating-position reanalysis against KPC-2, KPC-14, and KPC-159 references. Question: at the individual nucleotide positions that distinguish KPC-2, KPC-14, and KPC-159, how confidently do the deposited pre-treatment reads support a KPC-159-specific call versus a generic KPC-14-family call? Data: deposited reads plus the three public reference sequences (including OQ450354.1). Method: align reads to all three references and compare base-level support at discriminating positions rather than relying on whole-read best-match classification. Limitation: nanopore basecalling error rates at individual positions could themselves obscure a true single-nucleotide difference, so this reanalysis would need to report per-base quality alongside any conclusion.

Extensions requiring laboratory or clinical collaborators

3. A dilution-series validation with controls. Question: at what true minority-allele frequency does the CARD protein homolog model reliably detect a spiked-in resistance gene, and with what false-positive rate? Method: mix known ratios of a resistant and susceptible *K. pneumoniae* strain, sequence the mixtures, and compare detected versus true target-gene frequency, including negative controls and barcode cross-contamination checks. The target gene's copy number in each strain would need to be characterised, because bacterial mixing proportions do not necessarily equal resistance-gene proportions. This is a wet-lab study I have not conducted and would need a microbiology collaborator and access to characterised reference strains to attempt.

4. A prospective multi-patient comparison. Question: does nanopore-based resistance detection outperform routine diagnostics across a patient cohort, using standardised colony input and a pre-registered confirmation rule? Method: parallel sequencing and routine testing on consecutive eligible isolates, with turnaround time and diagnostic accuracy measured against appropriate reference standards — short-read sequencing to confirm gene and allele identity, and independent reference susceptibility testing, not short-read sequencing, to confirm phenotypic antibiotic susceptibility. This would require institutional clinical collaborators, ethical approval, and a patient population large enough to estimate sensitivity and specificity meaningfully — well beyond what I could undertake independently.

14. Questions the Paper Leaves Open

- Would the CAZ-AVI MIC strip — the relevant phenotypic comparator for CAZ-AVI susceptibility, since the VITEK2 AST-GN69 card did not itself measure CAZ-AVI MIC — report a different result on a colony pool closer in size to the sequencing input? The Methods do not specify a colony input for the lateral-flow carbapenemase assay, so its comparison to sequencing input remains uncertain regardless of what a matched-input experiment on the MIC strip might show.

- Is the single pre-treatment blaKPC-14-labelled read genuinely KPC-159, or a related-but-distinct KPC-family variant that happens to align closely? Short-read confirmation was only performed on the post-treatment isolate, so this remains unresolved.
- How much of the 0.56%-to-26.6% normalised abundance change reflects true within-patient enrichment under CAZ-AVI selection, versus differences introduced by separate culturing, extraction, and sequencing runs of two isolates collected a week apart from different body sites?
- Would a second technical replicate of the same pre-treatment isolate reproduce the same detection timeline shown in Figure 3, or does that curve reflect some run-specific variability?
- Is the predicted conjugative machinery on the IncN plasmids sufficient to infer that horizontal transfer actually occurred in this patient, or only that it was structurally possible?
- How would this workflow's accuracy and timing change on direct clinical specimens, where background human and microbial DNA would dilute the target pathogen's signal relative to a pure cultured isolate?

15. Lessons Learned

The clearest lesson from this paper is that a single well-chosen case study can carry real scientific weight without needing to carry more than it should. The pre-treatment single-copy detection, followed by its confirmation via a separate 8-hour run on a resubcultured technical replicate, is a genuinely interesting demonstration of an adaptive sequencing philosophy — keep sequencing until you have enough evidence — that stands on its own. It did not need the added framing about surpassing clinical practice to be worth publishing, and reading past that framing to find the narrower, better-supported claim underneath was the most useful exercise this analysis involved.

A second lesson is more procedural: this article taught me to pay close attention to which sample a downstream confirmation step was actually applied to. It would have been easy to read “KPC-159 was confirmed” and “the pre-treatment isolate carried a single KPC-14-labelled copy” as describing the same, fully verified fact, when in reality only the post-treatment isolate went through reference-laboratory short-read confirmation. Tracking exactly which claim attaches to which sample, rather than which claim attaches to the paper as a whole, turned out to matter more here than in most other papers I have read for this portfolio.

Third, working through the three different “copy number” quantities in Section 7 sharpened something I had underappreciated before: that a depth-derived estimate, a normalised ratio, and a raw detection count can share vocabulary while measuring genuinely different things, and that collapsing them into one number in a summary would misstate what the study found.

16. Connection to My Development as a Bioinformatician

This case study sits closer to clinical microbiology than most of what I have worked through so far in this portfolio, and it pushed me to get more precise about a distinction I had mostly encountered in the abstract before: the difference between a bioinformatics pipeline producing a confident-looking number and that number being clinically actionable. A 93% detection accuracy from the CARD protein homolog model is a statement about how the model scored a match against its own reference database; it is not, by itself, a statement about how often a patient with that detection actually has resistant disease. Keeping those two kinds of confidence separate feels like exactly the habit I want to build before I am the person producing pipeline output that a clinician or wet-lab collaborator might read past its caveats.

Working through the plasmid-level analysis — copy-number estimation from sequencing depth, MOB-suite's typing of mobilisation genes, the conjugation prediction — was also useful groundwork for thinking about resistance genes as sitting on mobile genetic elements rather than as isolated markers. That framing, that a resistance gene's

plasmid context (copy number, mobility, co-located genes) shapes how it spreads and how stably it persists, is one I expect to come back to in future reading, and this was the first paper in my portfolio where that context was central to the argument rather than a side detail.

17. Conclusion

Sequencing detected a low-abundance blaKPC-14-labelled signal in the pre-treatment isolate despite its reported CAZ-AVI susceptibility. A separate 8-hour sequencing run on a resubcultured technical replicate provided additional support. In the post-treatment isolate, sequencing identified a resistance-associated gene signal despite a negative carbapenemase assay result; routine phenotypic testing had detected CAZ-AVI resistance. That finding is worth taking seriously as an argument for genomic surveillance and for adaptive, extend-as-needed sequencing strategies in complex infections.

It is a narrower finding than the article's Discussion sometimes suggests. This was a retrospective, single-patient case study performed on cultured isolates after the clinical course had already concluded; the initial pipeline labelled its detection blaKPC-14 rather than identifying the truly novel blaKPC-159 subtype, which required a separate short-read confirmation step applied only to the post-treatment isolate; in the post-treatment isolate, the routine workflow's gap was in carbapenemase detection, while phenotypic testing correctly identified CAZ-AVI resistance; and the comparison between methods was not matched for colony input. None of these points undercuts the core detection result. They do mean that claims about superior diagnostic accuracy, validated clinical thresholds, or improved patient outcomes remain proposals for future work rather than results this study has already delivered. Read as a proof of concept with a clearly marked path toward validation, rather than as a validated diagnostic advance, this is a useful and honestly reported piece of work.

18. References

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